

5.- Determinar una ecuación paramétrica de la recta L que pasa por el punto $A(-4, -1, 0)$ y es paralela al eje z .

$$\vec{u} = (0, 0, 1)$$

$$L: \begin{cases} x = -4 \\ y = -1 \\ z = t \end{cases} \quad t \in \mathbb{R}$$

6.- Sea la recta R de ecuaciones

$$R: \begin{cases} 2x - y + 5 = 0 \\ x + y + 3z - 14 = 0 \end{cases}$$

Determinar una ecuación cartesiana en forma simétrica de R .

$$S: \boxed{z = 0}$$

$$2x - y = -5 \dots \textcircled{1}$$

$$x + y = 14 \dots \textcircled{2}$$

$$3x = 9$$

$$\boxed{x = 3}$$

$$A(3, 11, 0)$$

$$\begin{aligned} 3 + y &= 14 \\ \boxed{y = 11} \end{aligned}$$

$$S: \boxed{x = 0}$$

$$B(0, 5, 3) \rightarrow \vec{b} = (0, 5, 3)$$

$$\boxed{y = 5} \dots \textcircled{1}$$

$$y + 3z = 14 \dots \textcircled{2}$$

Sust. "y" en $\textcircled{2}$

$$5 + 3z = 14$$

$$3z = 9$$

$$\boxed{z = 3}$$

$$R: \left\{ \frac{x-3}{3} = \frac{y-11}{6} = \frac{z}{-3} \right\}$$

$$\vec{u} = \vec{BA} = \vec{a} - \vec{b}$$

$$\vec{u} = (3, 11, 0) - (0, 5, 3)$$

$$\vec{u} = (3, 6, -3)$$

7.- Calcular la distancia del punto $Q(2, 3, 6)$ a la recta L de ecuaciones

$$L: \begin{cases} x+2=7-y \\ z+2=8 \end{cases}$$

$$d = \frac{|(\vec{q}-\vec{p}) \times \vec{u}|}{|\vec{u}|}$$

$$L: \frac{x+2}{1} = \frac{y-7}{-1} = \frac{z+2}{4}$$

$$P(-2, 7, -1) \rightarrow \vec{p} = (-2, 7, -1)$$

$$\vec{u} = (1, -1, 4)$$

$$d = \frac{|(2, 3, 6) - (-2, 7, -1) \times (1, -1, 4)|}{|(1, -1, 4)|}$$

$$d = \frac{|(4, -4, 7) \times (1, -1, 4)|}{\sqrt{1^2 + (-1)^2 + 4^2}}$$

$$d = \frac{|(-9, -9, 0)|}{\sqrt{1^2 + (-1)^2 + 4^2}} = \frac{\sqrt{(-9)^2 + (-9)^2}}{\sqrt{1^2 + (-1)^2 + 4^2}} = \frac{\sqrt{81+81}}{\sqrt{18}}$$

$$d = \frac{\sqrt{162}}{\sqrt{18}} = \sqrt{\frac{162}{18}} = \sqrt{9} = 3 \text{ u.l.}$$

$$\vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -4 & 7 \\ 1 & -1 & 4 \end{vmatrix} = -16\vec{i} + 7\vec{j} - 4\vec{k} - (-4\vec{k} - 7\vec{i} + 16\vec{j}) = -16\vec{i} + 7\vec{j} - 4\vec{k} + 4\vec{k} + 7\vec{i} - 16\vec{j} = -9\vec{i} - 9\vec{j} = (-9, -9, 0)$$

8.- Obtener el ángulo agudo que forman las rectas L y R expresadas por

$$L: \begin{cases} \frac{2x-3}{4} = \frac{5-z}{2} \\ y = -3 \end{cases}$$

$$R: \vec{r} = (2-3t, 4, -t) \quad t \in \mathbb{R}$$

Sim

$$L: \begin{cases} \frac{x-3/2}{2} = \frac{z-5}{2} \\ y = -3 \end{cases} \rightarrow \vec{u} = (2, 0, -2)$$

$$\vec{v} = (2-3t, 4, -t) \quad t \in \mathbb{R}$$

$$\rightarrow \vec{v} = (-3, 0, -1)$$

$$\theta = \arccos \frac{|\vec{u} \cdot \vec{v}|}{|\vec{u}| |\vec{v}|}$$

$$\theta = \arccos \frac{|(2, 0, -2) \cdot (-3, 0, -1)|}{\sqrt{2^2 + (-2)^2} \sqrt{(-3)^2 + (-1)^2}}$$

$$\theta = \arccos \frac{2(-3) + (-2)(-1)}{\sqrt{8} \sqrt{10}}$$

$$\theta = \arccos \left(\frac{-4}{\sqrt{80}} \right)$$

$$\theta = 116.6^\circ$$

$$\theta = 180^\circ - 116.6^\circ$$

$$\theta = 63.4^\circ$$

Nombre:

Día

Mes

Año

Folio

Tema:

9.- Determinar si las S y T son perpendiculares, paralelas o coincidentes.

$$S: \begin{cases} x = 3 + m \\ y = -1 - 2m \\ z = 4 - m \end{cases} \quad m \in \mathbb{R}$$

$$\hookrightarrow P(3, -1, 4) \rightarrow \bar{p} = (3, -1, 4)$$

$$T: \bar{t} = (2 - 2n, 1 + 4n, 5 + 2n) \quad n \in \mathbb{R}$$

$$\hookrightarrow \bar{v} = (-2, 4, 2)$$

Iguando P , a la 1^{a} comp. $\bar{t}$

$$3 = 2 - 2n$$

$$2n = 2 - 3$$

$$2n = -1$$

$$n = -\frac{1}{2}$$

Sust n en $\bar{t}$

$$\bar{t} = (2 - 2(-\frac{1}{2}), 1 + 4(-\frac{1}{2}), 5 + 2(-\frac{1}{2}))$$

$$\bar{t} = (2 + 1, 1 - 2, 5 - 1)$$

$$\bar{t} = (3, -1, 4) = \bar{p}$$

$\therefore S, T$ coincide